EDGE DEVICE MEASUREMENT REPORT

Electronic Nose Coffee Classification System

Platform: NVIDIA Jetson Orin Nano

Model: Long Short-Term Memory (LSTM)

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MEASUREMENT SUMMARY

Single-Sample Inference Latency	Avg Total Power (Training)	Training Time
2.594 ms	**9.09 W**	**113.11 min**

MODEL CONFIGURATION

Model Type	Long Short-Term Memory (LSTM)
Epochs	50
Input Shape	8 sensors x 300 timesteps
Architecture	Recurrent Neural Network (LSTM)
Single-Sample Inference Latency	2.594 ms
Model Size	349.90 KB
Number of Parameters	642
Memory Footprint	6448.00 MB
Power Measurement	9.09 W

POWER, THERMAL & RESOURCE MEASUREMENTS DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	9.09 W	**Avg CPU Temp**	54.0 C
Peak Total Power	11.37 W	**Peak CPU Temp**	61.8 C
Avg CPU+GPU Power	3.85 W	**Avg GPU Temp**	53.4 C
Peak CPU+GPU Power	5.93 W	**Peak GPU Temp**	60.7 C
Avg RAM Usage	4615 MB	**Avg GPU Util**	42.9 %
Peak RAM Usage	6448 MB	**Peak GPU Util**	99.0 %
Energy Used	59648.0 J (16.5689 Wh)	**Training Duration**	113.11 min

GAS SENSORS USED (8 Sensors)

TGS2600	**MQ135**	**MQ3**	**MQ6**
MQ7	**TGS2602**	**TGS2611**	**TGS2620**